

&BEYOND NIGHT DRIVES

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Remember: the greatest challenges of the NIGHT DRIVE:

- To create the right atmosphere that suits this amazing wilderness.
- To interpret and communicate well - the Guests can't see at night and are therefore relying on you to PAINT THE PICTURE!

1. EQUIPMENT

- Small torch (fully charged or carry spare batteries) - for vehicle breakdowns and shining on visual aids.
- Pit stop bag, including insect repellent - for Guest peace of mind.
- Spotlight and red filter.

2. DRINKS BREAK

- Depending on when the night drive takes place, this will be a very important part of the drive - for a leg-stretch, a smoke, and the peace and scenery of the evening.



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- Choose a good area which is safe for you and the Guests and, if possible, which will provide good photographic opportunities - everyone loves to take sunset photos!
- Quickly let Guests know and select a suitable and SAFE place for a toilet break - if you don't show them, they will make their own plan when you are not looking!
- AWESOME HOSTING - this is an important part of your job as a guide - to read the Guests and set the scene: good slick service, knowledge of the drinks you have and the drinks you can make! (e.g. don't forget the lemon when you offer a G&T!)
- Unless it is a party, keep things relatively quiet - the sense of expecting something is better that way - listen for the sound of lions or leopard as the evening sets in.
- This can be a good time to start with the stars and planets - Venus, Jupiter are often visible quite early and are a useful starting point for looking at the skies.
- Watch for Guests, especially women that don't get off - it usually means that they need the toilet, but don't want to go!
- Prepare and test the spotlight long before you start packing up to go - you at least have a chance to fix or call for help before it impacts on the Guests.
- After drinks, check that no Guests want to go directly back to the lodge - some Guests don't enjoy night drives and rather want to have a long, relaxing drink, shower etc. before dinner (at lodges where afternoon and night drives are done together).
- Cross-selling: use the opportunity to talk about other lodges. esp. for Guests who are keen on stars, talk about the rooms with skylights, the telescope and resident astronomer at Sossusvlei Desert Lodge in the Namib Desert.

3. NIGHT DRIVE INTRODUCTION

- Where are you going/orientate the Guests.
- What are you looking for? What are you going to do? How long will you be out? (roughly).
- Explanation of the tracker's role (spotting and occasionally looking down for tracks of the larger predators).
- Talk about not shining on diurnal animals and a brief explanation of the use and working of a red filter - very important for their expectations and your sanity! (Guest: there, there, eyes, on the right, no, on the left, there, there, what's that...).
- Tell Guests to relax, look at the stars and to breathe in the different smells etc. (if not too cold, consideration must be given to taking off the canopy - if not in a National Park).
- Night drive intro:
 - There's a whole different world of animals that are active after dark, and it's these animals we're looking for now.
 - Mention some of the animals you could expect to see in your area. You should have done this at tea, so it's just a chance to remind your Guests.
 - As we're driving, (Tracker's name) will sweep the spotlight from side to side, looking for the light's reflection in animals' eyes.
 - If he spots anything, he'll signal to us and we can stop and have a look - don't try and follow the light, because you'll get a sore neck!
 - If we stop to watch any animal, (Tracker's name) will put a red filter over the spotlight. This takes the bright glare off the light and is much easier on the animals' eyes, which means they're more likely to relax and carry on about their business.
 - Red is also the colour that an animal's eye is least sensitive to in low-light conditions, which means we're minimizing our impact on the animals.

- Very importantly - we may come across some of the diurnal / daytime animals, and we want to leave them alone. This spotlight can blind them temporarily and put them at much more risk than they need to be at night. So, if (Tracker's name) spots any of them, he'll just sweep right past - don't worry about him missing anything!
- We'll take an easy drive back to the lodge and see what we can find. Shall we stop along the way for a little star-gazing? (OR deliver a brief and coherent outline of the drive's plan, explaining necessary sensitivities.)

4. SIGHTINGS PROCEDURE @ NIGHT

- SENSITIVITY is the most important thing - we are being allowed a privileged view into the lives of the night creatures. They must trust us!
- Following distance of mobile animals like lion, leopard: be sensitive, usually when the animals move off the road or look back over their shoulders - you are TOO CLOSE. You are disturbing them and might lose the sighting! Also, use red filter to view the animals (and see the road), NOT the vehicle's headlights.
- Cheetah - not to be viewed at night: explanation given to Guests as to their vulnerability to other predators.
- Elephant - ideally not viewed at night. You are unable to see the whole herd and keep watching all the animals. Elephants too are often unhappy with the spotlight. If you're in an open area and the elephant are not too close, switch off and listen in the dark - a great effect! Moonlit nights are great for elephant experiences.
- Hunting animals - this will always need to be managed carefully. Red filter is critical here. Read the situation, explain to Guests, keep the lights and engine off. Check with the red light every 5-10 minutes. The key here is the listening experience until the kill is made - only then can we reposition and view the animals with the red filter. We MAY NOT influence the outcome of a hunt under any circumstances. If a female predator has cubs, the sighting should be closed and the animal left undisturbed.
- Parking: if more than one vehicle is in a sighting, ensure that vehicles are parked more or less on one side - better for the animal and also the Guests who will otherwise be blinded!
- Share the spotlight duties between ranger and tracker or alternate rangers - this will help to change the Guests' views of the animal and also reduce insect bother and spotlight heat for one person holding the light!! Only one spotlight on the animals at a time!

5. ADDITIONAL THINGS TO DO @ NIGHT

- Frogging - in private or concession areas where safe to do it!
- STARS - very important to have basic stars under the belt - try for an African angle, local cultural things - STORIES - everyone loves a story. You have to create the opportunity and atmosphere to look at the stars.
- Spiders - for only a few interested people! (In areas where you get out on foot)
- Driving with the lights off in a full moon! An amazing different touch - be very careful - only where you know the road well and where there is no chance of encountering elephants or hanging branches.

SAFETY: being off the vehicle at night, snakes, leeches, mosquitoes and venomous spiders and scorpions must be taken into consideration. Use your discretion well! You are fully responsible. The

tracker should stay on the vehicle and scan with the spotlight periodically to ensure the area remains safe.

6. SPOTLIGHT ETIQUETTE

IN (the right thing to do)	OUT (don't do it!)
Use of a small torch as an alternative to the spotlight when viewing small things and in sensitive sightings	Tracker shining in a wide arc beyond 180° that shines in the eyes of Guests and the ranger.
Using a spotlight to point out the stars. Even better, use a green laser-pointer	'Holding' an animal in the centre of the spotlight, use softer light if possible.
Using a spotlight at a low angle to point out and accentuate fresh tracks on the road	Shining 'wishfully' at eyes in the bush, usually blinding antelope.
USE RED FILTERS! Be quick to put in on the spotlight!	Shining on known roosts of waterbirds and others that are not adapted for night flight! Or shining on roosting birds too soon after sunset when they are not yet settled and sleeping
Use of a small torch or spotlight to point out pictures of nocturnal animals in field guides	Not dimming your headlights when approaching herds of antelope or birds on the road

EXCEPTIONS TO THE RULES:

- If a predator is too close and either you or the Guest is feeling uncomfortable, you can, under these special circumstances, shine the spotlight directly into its eyes - this will be uncomfortable for the animal and will usually turn it away from you. This does NOT work for elephant, though, and they may try and eliminate the irritating light, so simply keep a safe distance in the first place!

7. NOCTURNAL CREATURES

As a guide you need to get good, interesting and relevant information on these creatures and understand their place and role in the ecosystem etc. Guiding techniques must be developed to view/listen to them in an exciting and yet sensitive manner.

Below is a list of creatures which can be viewed on a night drive, some of which are fully nocturnal and others which are both diurnal and nocturnal:

NOCTURNAL MAINLY	NOCTURNAL AND DIURNAL (BOTP)
Bats	Chameleon - actually, diurnal, but you find them easier at night!
Bushbabies (Galagos)	Leopard
Viverridae (genets, polecats, civets, mongooses and weasels)	Lion
Moths (esp. emperor, hawk and moon moths)	Small cats - serval, caracal, African wild cat
Porcupines	Hyena and jackals
Hares	Crocodile

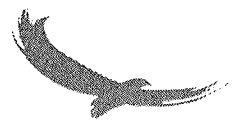
Owls, nightjars, thickknees and coursers	Honey badger
Aardvark	Pangolin
Aardwolf	
Micromammals (rodents and shrews)	

Hippopotamus and grey duiker are primarily nocturnal in their foraging, although we DO NOT shine the spotlight on them (even with a red filter) for sensitivity reasons. It is possible to view both during the day.

8. INSECTS, SPIDERS AND MALARIA

Knowledge of potentially harmful insects is very important. Using a spotlight attracts insects and with the millions of different types, anything could land in the vehicle or on a Guest!

- Let them know not to crush or slap insects, but to flick them away. This is important for **blister beetles** which will cause severe skin burning if crushed. The European equivalent of this is the Bombardier Beetle which many Guests will know.
- In summer especially, flying **dung beetles** can cause pain and damage when flying at high speed into someone's eyes.
- Spotlight management when there are termite eruptions and other insects gathering must be good to avoid Guests (and you!) feeling great discomfort. Generally, people do not like insects!
- Remember though that a **termite** eruption is one of the best sources of action in the African bush. All sorts will come to feed. You need to know termite ecology well - predators, life-cycle etc. Sensitivity to Guests and animals trying to feed is very important.
- Basic knowledge of the life-stories (so you can tell a story) of the following insects is important: Fireflies, adult antlions, egg-laying mantids and katydids
- **SPIDERS!!** You are almost guaranteed to see these in the summer/wet season. The larger **Golden Orb Spiders** and **Bark Spiders** spin large webs that often go across the road. Sensitivity is required - to Guest and spider. Many Guests will have a genuine fear (arachnophobia), and others might be fascinated by seeing a real, wild spider spinning its web. Many night drives have been made by guides getting excited and interpreting this event well. On some night drives, this could be all you see! When Guests have the fear, just move past quickly and leave the spider sighting alone.
 - When you do find this web, what do you do if it blocks your path? If there is another easy road to take, take it. If not, then try and gently break the bottom strand, giving the spider time to move away to a side branch. If you do it too quickly, she will get entangled in her own web! Driving around might be necessary if Guests are really scared, but the off-road driving is usually not necessary.
 - Remember to explain to your Guests what you are doing and why.
 - Wolf spiders are often easy to point out as you will see their eyes reflecting from where they are on the road.
 - Community nest spiders are nocturnal and this is the time to get lucky and see the colony in action.
- Questions for you to read up on:
 - How do fireflies produce the green light and why don't they overheat?
 - What is silk made of and how does it work?
 - What can you say to Guests when they ask about malaria?



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9. SLEEP THEORY

What is sleep? A strange question, but when you look at the animal world, you notice that different animals sleep for different lengths of time - why? Why or when could you talk about it? You can spot sleeping birds and also some mammals if you're lucky.

- Is sleeping = no behaviour or is sleeping its own behaviour? A big question!
- Animals prepare for sleep, they shift position during sleep and have periods of yawning and stretching - this indicates a behaviour.
- The next big question: what is the purpose of sleep? To recover/recharge the battery or to switch-off/inactivate?
- In humans we tend to sleep to recover - but for how active we are we should then be the biggest sleepers of all (human average is only 8hrs per day)
- The 'Inactivation theory' - this sees sleep as the great immobilizer. Maybe sleep prevents animals from being worried about not doing something. Maybe it prevents animals from jumping up at every slight disturbance in the dark.
- Only animals that need to feed constantly to stay alive, or those that are very vulnerable day and night, will benefit from staying awake all the time - we see this: the **shrews** hardly sleep at all - only a few minutes at a time! **Antelope** sleep very little. Most **ungulates** take only a few hours each night.
- Some of the great sleepers: **Bats** - 19hrs, **Sloth** - 20hrs
- **Elephants** mostly sleep no more than 2 hours per night - their internal organs probably suffer under the great weight.
- **Giraffe** - 4hrs, but never more than 20 minutes deep sleep (with neck on rump) - this could be due to predation and/or blood pressure problems
- Birds and mammals **DREAM!** This is active sleep. Dreaming is possibly the brain trying to file all the difficult information from the day. It might also be a protection to keep the brain moving whilst sleeping. Remember that there is always huge muscle relaxation in deep sleep - it's not hard to think why this will be a problem for animals.
- Some birds sleep with one eye open - like **flamingos** - it is thought by some people that this allows one half of the brain to rest. It then changes the sleeping and alert side from time to time. This way they will be refreshed without ever having been unwary to danger. Only a theory. This is also possibly how **Swifts** are able to sleep whilst still flying high in the sky!

When you see things sleeping, try and interpret their behaviour and talk about some of the information above - it will make you and the Guests THINK!

10. NIGHT VISION AND RED FILTERS

- Different creatures have different abilities to see in the dark. This is related to the structure of the eye and the cells inside them.
- During bird migration time: They can use the stars to fly at night. Wind and clouds block them out and force them off course - you can talk about this in the change of season when the birds are migrating.

- Some birds like Owls 'see' with their ears as well and snakes active at night might 'see' with heat-sensitive pits on the upper lip. Bats use their ears and leafnoses to echolocate and 'see' that way.
- Oilbirds (relatives of the nightjars living in caves in South America) don't see at all but echolocate like bats!
- Eye-shine: What creates the eyeshine? A special layer called the *Tapetum lucidum* - this reflective layer at back of the retina basically bounces light back into the eye, increasing the total amount of light available to the eye - to see better. The eyeshine is the photons/light reflecting back.
- You can make some generalizations about the colour of the eyeshine from certain animals, but the colour will often depend on the angle that the spotlight shines the light into the eye.
- Strangely although nightjars have strong eyeshine all the same colour - orange/red, very few owls exhibit eyeshine.

Why Red Filters on Spotlights?

We avoid spotlighting animals which are not strictly nocturnal only in an attempt to reduce the overall spotlight impact we have, not because nocturnal or crepuscular species are less sensitive to light or that diurnal species are more sensitive. You must be able to explain this concept to Guests - the effect of light is the same on the eyes of all animals. The reason we do night drives is to try and see animals that we are unlikely to get a chance to see in the day!

Many animals commonly seen during the day are active at night and are still reliant on their excellent night vision ability for success as any other animal. Red filters help reduce the impact we have on those animals which we choose to view after dark. It has been clearly shown that many animals are less disturbed by red light and that using red filters minimizes the effect that the spotlight has on their night vision sensitivity (this is not true for blue light).

Of the light-sensitive cells of the retina at the back of the eye, CONES produce detailed, colour images and require bright light to be activated. RODS produce less detailed images and are far more sensitive in low light conditions and very sensitive to movement.

To check: at night, stare directly at an object - not easy to see. Look at it through the side of the eye - you can often make out shapes and some detail. The saying that you saw something out of the corner of your eye is not for nothing!

The eye has basically 2 ways of adjusting to changing light conditions:

As the light drops,

1. the pupil dilates (gets bigger) to let in more light and ...
2. the retinal tissue adjusts chemically to make the rods more sensitive to light or 'fully sensitized' (the Rhodopsin cycle)

As the light increases from dark,

1. the pupils contracts (gets smaller) to let in less light and...
2. the retina tissue reverses its chemical process to desensitize the rods until the eye is 'fully light adjusted' or desensitized.

The full range of adjusting that the eye can do is measured as 9 log units. Significantly, the pupil can only effect 1 log unit of change, but the retina can effect 8 log units!



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So, although a lion's pupil may shrink to a pinprick size in the full spotlight, it is only able to reduce the light to the eye by a small percentage. The rest of the change must happen in the retina by chemical process.

In simple terms: it takes about 30 minutes for eyes to become fully sensitized in low light (i.e. to get the 'night vision'). This sensitivity is lost very quickly (1-2 minutes) when exposed to bright, white light. Even when you shine a spotlight away from the animal against a bush or shine with half a red filter only, white light is reaching the eye and the 'night vision' is being lost quickly.

HOW DOES THE RED FILTER ACTUALLY WORK?

The ROD cells in the retina are very sensitive to BLUE light (this is contained in white light). When exposed to blue, the rods are desensitized quickly - 'night vision' is lost - even staring at the moon can have this effect! RED light does not have this effect on the rods and that is why we use the filters - to block out blue and let through red.

IS THE RED LIGHT INVISIBLE TO THE ANIMALS?

No. Most people agree that they can see more contrasts in the red light. In other words, the shadows from the animal will be more visible to the other animals around it. Therefore, red filter doesn't allow us to shine on hunting cats etc.

By using red filters, we are simply having much less impact on animal's night vision sensitivity and so less impact on their lives and interactions. Over time we can improve the length and quality of sightings too, especially the smaller nocturnal creatures.

11. SMELLS AND SOUNDS

It is as in all your game drives important to engage the other senses of your Guests for the full experience.

- Just after sunset, the smells are more intense in the still air. Grasses, Potato Bushes (*Pyllanthus*), Jasmine-type smells like Num-nums, Gardenias, Jasmines and even some orchids - great to smell.
- These also provide a great link to moth pollination.
- Carcass **smell** - without disgusting Guests, these can be smelt briefly - it is real and will often stimulate discussion! As the Guests smell you can talk about the smell sense of animals like hyena. Remember that lion and leopard don't really have excellent smell sense - they rely more on vision.
- Stop when you have a chance to **smell** fresh scent-marking by animals like genet, civet and leopard. When you are following a territorial animal and you see it scent-marking, slow down to smell as you go past.
- Take time to **listen**. Show your Guests the listening technique of cupping hands behind ears - they will always be amazed.
- **Listen for:** frog chorus, elephants, herbivores grazing nearby, nightjars calling and the distant roaring and coughing of lion and leopard. Sounds of nearby water.

Touching and tasting: Guests will generally be more comfortable to do this in the daylight!



12. GENERAL SENSITIVITY

- Careful driving at night - it is easy to chase off slow moving snakes like python if you are not paying attention.
- Do not drive at more than 30kph - it is easy to kill birds on the road and it is cold!
- Toilet breaks - have your ears open to listen to Guests - often when they go quiet, they need the toilet but are too shy to ask - look for simple opportunities to offer this. A full bladder will ruin the experience for the Guests.
- Use of bird call recordings to call up owls - not ideal, but can be done to good effect occasionally for very interested, birder Guests.

Overall: remember that even for you as the guide, the night drive will be what you make of it - an adventure into the unknown or a boring waste of time in the dark. Choose the first option!